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STUDY PROTOCOL

Data management protocol for healthcare services research using administrative data: Indigenous data governance and sovereignty, Anas Wayuu, Colombia

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Abstract
The governance of health data is an essential component of territorial health care management and the design of public policy. Proper governance enables the strengthening of data management capacities throughout the data lifecycle. The examination of health data governance is of particular relevance in relation to Indigenous Peoples, where the concept of Indigenous data sovereignty has gained traction over the last 30 years. The paper presents a data management protocol developed and implemented for a research study by an Indigenous health insurance organization from Colombia, Anas Wayuu. We describe and analyze the rationale behind technical criteria for processing and transferring administrative data; and document elements of a data management process for the structuring of data for the purposes of studying health care use within the context of culturally adapted health services. While the protocol was developed and implemented in partnership with outside academic partners, we showcased how it was done consistent with Indigenous data governance and sovereignty principles. To some extent the development and implementation of the protocol was aligned with the Findable, Accessible, Interoperable, and Reusable (FAIR) principles. However, Anas Wayuu being an Indigenous health organization had to consider them in relation to the rights over their own data, as per the CARE Principles for Indigenous Data Governance. While aspects of the two sets of principles can be somewhat complementary, our paper suggests a key tension.

Keywords: Indigenous data sovereignty, data governance, Indigenous healthcare, Wayuu, CARE, FAIR.

Abstract in Español at the end of the article

INTRODUCTION

The governance of health data is an essential component of territorial health care management and the design of public policy. Proper governance enables the strengthening of data management capacities throughout the data lifecycle. The process of data governance includes the: design of data use protocols; definition of resources; infrastructure maintenance; design of data analytics processes; and identification of the human resources and roles [1]. The examination of health data governance is of particular relevance in relation to Indigenous Peoples, where the concept of Indigenous data sovereignty has gained traction over the last 30 years, defined as "the right of Indigenous Peoples to own, control, access and possess data that derive from them, and which pertain to their members, knowledge systems, customs or territories" [2]. Our article showcases a data management protocol of a Colombian Indigenous health insurance organization, Anas Wayuu, that was developed for research purposes demonstrating Indigenous

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data sovereignty in action. We do so by describing the data governance and management approach developed by Anas Wayuu in accordance with Indigenous data sovereignty principles. Furthermore, we use a specific research study that assessed the impact of the intercultural hostels on healthcare access and that was conducted in partnership with academic partners, as an example of how this data management structure was applied in practice. The study assessed healthcare access of enrollees with a breast cancer diagnosis, pregnant persons and newborns with specific morbidities, and children under five years of age with acute malnutrition. For the purposes of the present protocol paper, we primarily report on the component of the study related to data management outcomes of children under five years of age with acute malnutrition.

Colombia's universal health system is organized in publicly and privately administered insurers [3] [Entidades Promotoras de Salud (EPS)] that are responsible for enrolment, provider contracting, and access to a nationally defined benefits package [4,5]. Colombian Indigenous Peoples have leveraged these reforms to establish Indigenous-led insurers and delivery entities [6,7]. Anas Wayuu is an Indigenous-led, non-profit insurer established in 2001, created by two Indigenous associations representing 120 Wayuu communities, currently providing coverage to over 295,000 individuals, 67% of whom are Indigenous Wayuu. The organization is governed by representatives of Wayuu communities, and is staffed largely by Wayuu human resources. It coordinates care through a contracted network of 27 Indigenous and 65 non-Indigenous health service provider institutions across public and private sectors, known as Health Provider Institutions (Instituciones Prestadoras de Salud - IPS), spanning primary to tertiary care [8].

Colombia has made progress in the design and implementation of knowledge management strategies under the integration and use of data to monitor health policy, obtain evidence on health outcomes and social inequities, and understand how it is situated in the regional and global context [9]. The Colombian government's intention to strengthen public health surveillance seeks an alignment with international data management standards for health research and public policy evaluation. In 2001, the Colombian Ministry of Health and Social Protection established the Health Services Provision Registry (Registro de Prestación de Servicios de Salud - RIPS) as an electronic data collection strategy to gather information on the use of services by the population that receives services in the national health system. RIPS have a standard structure for data collection, which is used by healthcare system stakeholders for management purposes. These records are the primary source of information for the management, regulation, and monitoring of the Colombian healthcare system. They contain patient identification data and information on healthcare service delivery to ensure the flow of information and facilitate financial exchanges between system actors. These records are also expected to be used to support decision-making in healthcare processes, to improve the administrative and operational management of the healthcare system, and for accountability purposes [10].

Anas Wayuu, being a health insurance organization, collects RIPS data from the health care providers it contracts with, enters it into its information system and sends a copy of the data to the Ministry of Health. Colombia has also implemented the Public Health Surveillance System (Sistema de Vigilancia en Salud Pública - SIVIGILA), a technological infrastructure that collects data on health conditions that are categorized as public health issues by the public health authority. The system includes protocols for various actors to generate, collect, validate, and store these data [11]. Anas Wayuu also collects and utilizes SIVIGILA data.

Among its programs, Anas Wayuu has implemented culturally adapted services to facilitate the interaction of Wayuu patients with healthcare services. One of these programs consists of a network of intercultural hostels that facilitates the stay of enrollees and companions in situations where they need to travel from their area of residence to urban centres for medical care [12]. In the interest of assessing the impact of the intercultural hostels on healthcare access, Anas Wayuu partnered with Universidad de Antioquia (Colombia) and University of Manitoba (Canada) researchers to jointly conduct a study using their administrative data. The study was conducted using data from 2014 to 2019 inclusive. The study's objectives were to explore health service utilization of Anas Wayuu enrollees that used intercultural hostels and that fell into one of the following categories: had received a diagnosis of breast cancer; pregnant persons and newborns with specific morbidities; children under five years of age with acute malnutrition. The research team (composed of Anas Wayuu and university-based researchers) utilized the experience and technical recommendations of a Canadian population research centre, the Manitoba Centre for Health Policy [13], for developing data management protocols to evaluate health programs.

The aim of paper is to showcase how data governance by an Indigenous health organization enables research following Indigenous data sovereignty principles. We do so by presenting the research agreement that was aligned with these principles; describing the rationale and technical criteria used to process and transfer administrative data to study service use of culturally adapted health services provided by Anas Wayuu; and documenting the outcome of the data management strategy.

Indigenous data governance and data sovereignty

The International Indigenous Data Sovereignty Interest Group formulated the "CARE Principles for Indigenous Data Governance" [14], meaning: that Indigenous data must facilitate Collective Benefit for Indigenous Peoples, Indigenous Peoples' Authority to control and

govern such data, when working with Indigenous data there is a responsibility to nurture Respectful relationships with Indigenous Peoples, and Ethics in data practices is representation and participation of Indigenous Peoples, who must be the ones to assess benefits, harms, and potential future uses based on community values and ethics [14]. Central to the research partnership between Anas Wayuu and university-based researchers were the principles of Indigenous data governance and data sovereignty. Putting into practice these principles meant that Anas Wayuu leadership defined the study's objectives, administered access to the datasets, and had decision making control over data management and analysis processes. This extended to study reports and peer-reviewed manuscripts submitted for publication. To explain how this was operationalized, we list below the main clauses that governed the research, as per the agreement that was signed by Anas Wayuu, Universidad de Antioquia and University of Manitoba, mapping in brackets how they match with CARE principles [15]. The CARE principles arguably have some overlaps with the Findable, Accessible, Interoperable, and Reusable (FAIR) principles [16]. More so, CARE has also been presented as complementing FAIR [14], a notion that in the discussion section we critique by identifying a key tension point between both sets of principles: 1) Access to administrative data of Anas Wayuu, as well as surveys conducted, is solely for the purposes of this particular study (Authority of Control); 2) Access to, and use of the data, has to strictly adhere to the principle that Anas Wayuu (as organization) owns the data, controls it, regulates and decides when and how the data is accessed, and is the data holder (Authority to Control); 3) Anas Wayuu has final decision regarding which variables and data periods will be used (Authority to Control/Ethics); 4) At any time during the research project, Anas Wayuu has the right to deny access to the data and/or require researchers to cease using it (Authority to Control); 5) Anas Wayuu provides the data to researchers only after it has been de-identified (Ethics); 6) Researchers and all individuals participating in the study have to guarantee the confidentiality of the information and records provided by Anas Wayuu. No individuals or institutions other than those who were part of the agreement can have access to the study data (Ethics); 7) Dissemination of research results through publications and other means is agreed to be of the interest of Anas Wayuu and partners (Collective Benefit/Responsibility; 8) The publication of study results must be reviewed by representatives of the three partner institutions (Collective Benefit/Ethics; 9) When the publication is written in English or a language other than Spanish, the authors must submit an executive summary in Spanish for evaluation by the representatives of Anas Wayuu. If either party so requires, the authors must provide additional information in Spanish to facilitate the evaluation of the publication (Authority to Control/Responsibility); 10) When differences arise between the researchers of Anas Wayuu, Universidad de Antioquia, and University of Manitoba in the interpretation of study results, each will have the right to explicitly state their position on the matter in dispute in the publication (Authority to Control/Responsibility); 11) If Anas Wayuu, with reasoned justification, does not fully approve a publication, it will not be published (Authority to Control); 12) The names of the individuals listed as authors of a publication will be defined for each publication on a case-by-case basis (Responsibility); 13) The research project ends on "DATE AGREED UPON". However, considering that the process of publishing results is lengthy, it is agreed that this agreement will be extended for a period of three years beyond the study's completion date, until "DATE" (Authority to Control). After the conclusion of the study, the data stays only with Anas Wayuu. Access to the data is controlled by Anas Wayuu information system staff, consequently researchers were unable to access the data after study conclusion.

The agreement was central to the obtainment of research ethics approval, which followed two steps. Ethics approval for the study proposal was first submitted to the Board of Wayuu Traditional Authorities that oversees Anas Wayuu. Second, once the approval from the Wayuu traditional authorities was obtained, the study was submitted to the University of Manitoba's Health Research Ethics Board. The approval of this board was necessary due to the involvement in the study of academic researchers from the University of Manitoba, which also covered Universidad de Antioquia academic researchers as they were under contract with the University of Manitoba.

METHODS

Data management

We followed recommended tasks for the management of socio-health data, including structuring and transferring the data, assessing its quality, and analyzing the data. The study consisted of a retrospective cohort design based on socio-health administrative data of individuals residing in La Guajira, who required or were offered healthcare services. The records of healthcare services of enrollees were captured by the providers (IPS) at the time of contact with services (Figure 1). The datasets included individuals' information related to their health insurance, use of healthcare services, provision of intercultural hostel services, and the surveillance and monitoring of events of public health interest (mandatory reporting by local health units of events of public health importance such as infectious diseases, environmental health risks, and so on). Records of their social and demographic characteristics were captured by Anas Wayuu either at the time of visits to the hostels, by notifications of SIVIGILA, or during access to healthcare services.

Data collection and assessment

For the study we used administrative health data sources for Anas Wayuu enrollees from the period 2014 to 2019. To identify cases of epidemiological surveillance events, we used data from SIVIGILA that includes clinical and demographic information on confirmed cases of Anas Wayuu enrollees suffering from one or more diseases. On the other hand, RIPS includes information related to the provision of healthcare services, such as hospital discharge summaries, emergency room visits, medical claims, prescriptions for medications and therapies, and birth registrations. RIPS also include identification and demographic information, including age, sex, area of residence, and user type [17]. Visitor records of hostels contain data on patients who use these facilities during the healthcare process. The characteristics include the date, location and duration of the hostel visit, the diagnosis of the primary illness, and the assigned procedure code. Data on high-cost illnesses include primarily cancer diagnoses and were considered exclusively for clinical confirmation of cases.

Data quality assessment was performed in two stages. The first stage was during data preprocessing and before data extraction from clinical cases and procedures, when we assessed data reliability in terms of data duplication and consistency. The second stage of data quality assessment was performed after data extraction but before data analysis [18]. Regarding the first step, the duplication indicator was calculated as the ratio between the number of records with at least one duplicate and the total number of records in the dataset. Figure 1 describes the outcome according to the dataset. RIPS clinical procedures and medication prescription records showed the highest duplication rates. Duplicate records were excluded from data extraction and analysis. Two main data extraction procedures were implemented to explore the use of health services among members suffering from any of the suggested health conditions. These are explained in the two sections that follow.

Framework review

First, we used national references from the Colombian Ministry of Health to develop a document review strategy considering the diagnostic, treatment, and rehabilitation processes of the natural history of the events. The document review strategy for each event involved six steps: 1) Identifying suggested clinical practice guidelines related to the events of interest; 2) Tracking regulations regarding services recommended by health authorities for the health care of the events; 3) Designing a matrix identifying the main elements of the diagnosis and treatment of each health condition; 4) Recording the recommended coding for diagnostic subcategories according to the 10th Edition of the International Classification of Diseases (ICD-10) and clinical procedures according to Colombia's Clasificación Unica de Procedimientos de Salud (CUPS) [Unique Classification of Health Procedures]; 5) Identifying the ICD-10 subcategories (four-digit codes) and CUPS codes recommended to record the diagnosis and treatment of the disease; 6) Validating the lists with experts and professional health service convened among the Anas Wayuu network of providers.

Data preprocessing

Second, data on health conditions, therapies and services accessed by these cases during disease management were extracted. The structuring of variable data for each individual was carried out through the design and implementation of an automated report using Structured Query Language (SQL). The query was implemented on processed data, excluding duplicate records. The primary diagnosis, date of care, procedure code, and patient identifier constituted the composite indicator of the records. Visits observed in each healthcare service were described according to the suggested coding standards (ICD-10 & CUPS) based on the identified healthcare services. For the calculation of statistical indicators, we used descriptive techniques such as absolute and relative frequencies, measures of central tendency, dispersion, and position. The suggested components for calculating quality indicators, such as rates, ratios, or proportions, had been previously tested in reports analyzing population data from public health in Colombia.

To implement the data extraction, transformation, and loading procedures, we used the open-source software PostgreSQL/PGadmin4. For statistical indicators, variable processing, quality validation, and results visualization, we relied on open-source resources from the R Integrated Development Environment. Final text and table editing was performed using applications offered by the Microsoft Office 365 platform.

Strict data privacy and confidentiality were maintained. All identifiable information was masked by Ana Wayuu's data administrator across all data sources used (i.e., no researcher had access to unmasked data). Records were not linked to other datasets outside of the study sources. Data de-identification involved removing personal identifiable information. For the purposes of the study, personal information was removed, and a random number was assigned as a unique identifier across all data tables implemented in the database. Access to the dataset was provided by Anas Wayuu's data administrator. Figure 2 provides a visual summary of the data management process.

OUTCOME OF THE DATA MANAGEMENT PROCESS

Access to administrative data included records from five subsystems comprising the data management process of the Colombian healthcare system. The use of microdata facilitated the exploration of missing values and duplicate records. Data quality outcomes showed similar patterns across subsystems. Insurance data suggested a significantly higher percentage of new members

Registration system	Health event (n variables)	2014		2015		2016		2017		2018		2019		2020		Total (N)
		Records (n)	Duplicate (n; %)	Records (n)	Duplicate (n; %)	Records (n)	Duplicate (n; %)	Records (n)	Duplicate (n; %)	Records (n)	Duplicate (n; %)	Records (n)	Duplicate (n; %)	Records (n)	Duplicate (n; %)	
Anas Wayuu (AW)	Hostel services (15)	54	(0; 0)	257	(5; 1.99)	807	(12; 1.51)	1,200	(10; 0.84)	1,544	(17; 1.11)	1,872	(27; 1.47)			**5,649**
Resources Manager (ADRES)	Health insurance enrollment (12)	139,870		10,982		9,069		60,050		17873		18,419				**256,310**
Individual records of health service provision (RIPS): Diagnosis, treatment and rehabilitation	Physician consultation (14)	141,887	(4,999; 3.72)	443,365	(18,633; 4.44)	628,618	(34,122; 5.78)	712,423	(8,997; 1.28)	850,928	(8,788; 1.05)	883,492	(12,177; 1.4)			**3,555,049**
	Births delivery (11)	761	(15; 2.01)	851	(14; 1.67)	1,105	(26; 2.41)	1,710	(21; 1.24)	1,583	(20; 1.28)	1,909	(18; 0.95)			**7,803**
	Hospitalization (16)	21,464	(218; 1.03)	25,431	(264; 1.05)	14,601	(220; 1.53)	17,012	(100; 0.59)	21,163	(425; 2.05)	21,859	(1,348; 6.57)			**118,941**
	Emergency visit (14)	24,246	(237; 0.99)	29,964	(249; 0.84)	32,870	(379; 1.17)	47,461	(340; 0.72)	58,046	(569; 0.99)	60,828	(1,963; 3.33)			**249,632**
	Clinical procedures (12)	332,727	(17,018; 5.9)	741,029	(38,210; 5.73)	917,746	(52,294; 6.32)	1,457,337	(61,029; 4.63)	1,832,987	(73,037; 4.47)	1,992,164	(114,587; 6.5)			**6,499,526**
High-Cost Accounts*	Ruinous diseases monitoring (4)									191	(2; 1.06)	178		710	(16; 2.31)	**914**
Public Health Surveillance System (SIVIGILA): Epidemiological surveillance*	Breast and cervical cancer (100)									17		13				**30**
	Low birth weight (143)									177		159				**342**
	Congenital syphilis (104)									5		5				**10**
	Acute, Moderate & Severe Child Malnutrition (112)									479		546				**1,062**
	Gestational syphilis (182)									54		53				**107**
	HIV/AIDS/ Mortality (139)									53		48				**103**
	Extreme maternal morbidity (151)									258		219				**487**

Source: Prepared by the authors based on data sources from the study
Prioritized health events: 1. Two population groups to explore maternal and child health events (1.1 pregnant women and 1.2 newborns), 2. breast cancer in the total population, 3. acute, moderate, and severe malnutrition in children under 5 years of age.
ID: identifier of the dataset in each registration system, AW: Indigenous Health Service Provider; n: absolute frequency.

Figure 1. Administrative health data sources for Anas Wayuu enrollees - 2014 to 2019.

in 2017 compared to the rest of the period. Similarly, the use of hostels showed higher frequencies starting in 2016. Regarding disease diagnosis and treatment data, the RIPS files had duplication rates of less than 10% in the years evaluated. The SIVIGILA data corresponded to notifications from 2018 and 2019. For instance, among the events reported by Anas Wayuu, 1,062 (2.5%) individuals were reported as being cases of childhood malnutrition in children under 5 years of age, of whom 47 were ruled out at the end of the care process. Of the total cases included in the study (n=2,141), 1,570 (73.3%) were confirmed cases with a single notification for each diagnosed event among members.

The elements used to extract data from patient groups and their interaction with health services were based on a review of clinical practice guidelines and national regulations. The coding of diagnoses and procedures suggested by the guidelines was grouped into four health event categories: breast cancer [19], pregnancy [20], newborn health and nutrition [21]. The recruitment, recording, and coding (according to ICD-10 and CUPS) of health services was defined by stages of care experienced by the individual: 1) Confirmation as a case reported to SIVIGILA among Anas Wayuu enrollees; 2) Access to health care for disease diagnosis, and 3) Access to services for treatment and/or rehabilitation. The ICD-10 subcategories were grouped into diagnostic blocks, where 40 were related to the breast cancer group, including family medical history and secondary prevention. For pregnant women and newborns, we selected 24 ICD-10 subcategory codes associated with infectious and chronic health conditions. For the nutrition group, 14 diagnostic subcategories for malnutrition and other nutritional deficiencies were documented. CUPS codes for disease treatment procedures were classified according to the affected body organ system or the diagnostic assistance. A total of 202 services were documented, with 113 related to breast cancer.

Based on the ICD-10 diagnostic block lists and CUPS code groups, we described the contact of cases with health services during the 2018-2019 period, its frequency, and its relationship to the care processes for each event. Events such as breast cancer reflected a greater use of outpatient services, health procedures, and hostel use, while among pregnant persons a higher frequency of hospitalizations and outpatient visits was observed. Among cases of maternal and newborn morbidity, demand for consultation and procedures exceeded 60% in three or more contacts with health services during the 2018-2019 biennium.

DISCUSSION

In the above sections we presented a data management protocol developed and implemented for a research study by a Colombian Indigenous health insurance organization, Anas Wayuu. We described and analyzed the rationale behind technical criteria for processing and transferring administrative data; and documented elements of a data management process for the structuring of data for the purposes of studying health care use within a context of culturally adapted health services. While the protocol was developed and implemented in partnership with outside partners, we showcased how it was done consistent with Indigenous data governance and sovereignty principles.

Our protocol is a contribution to a global demand

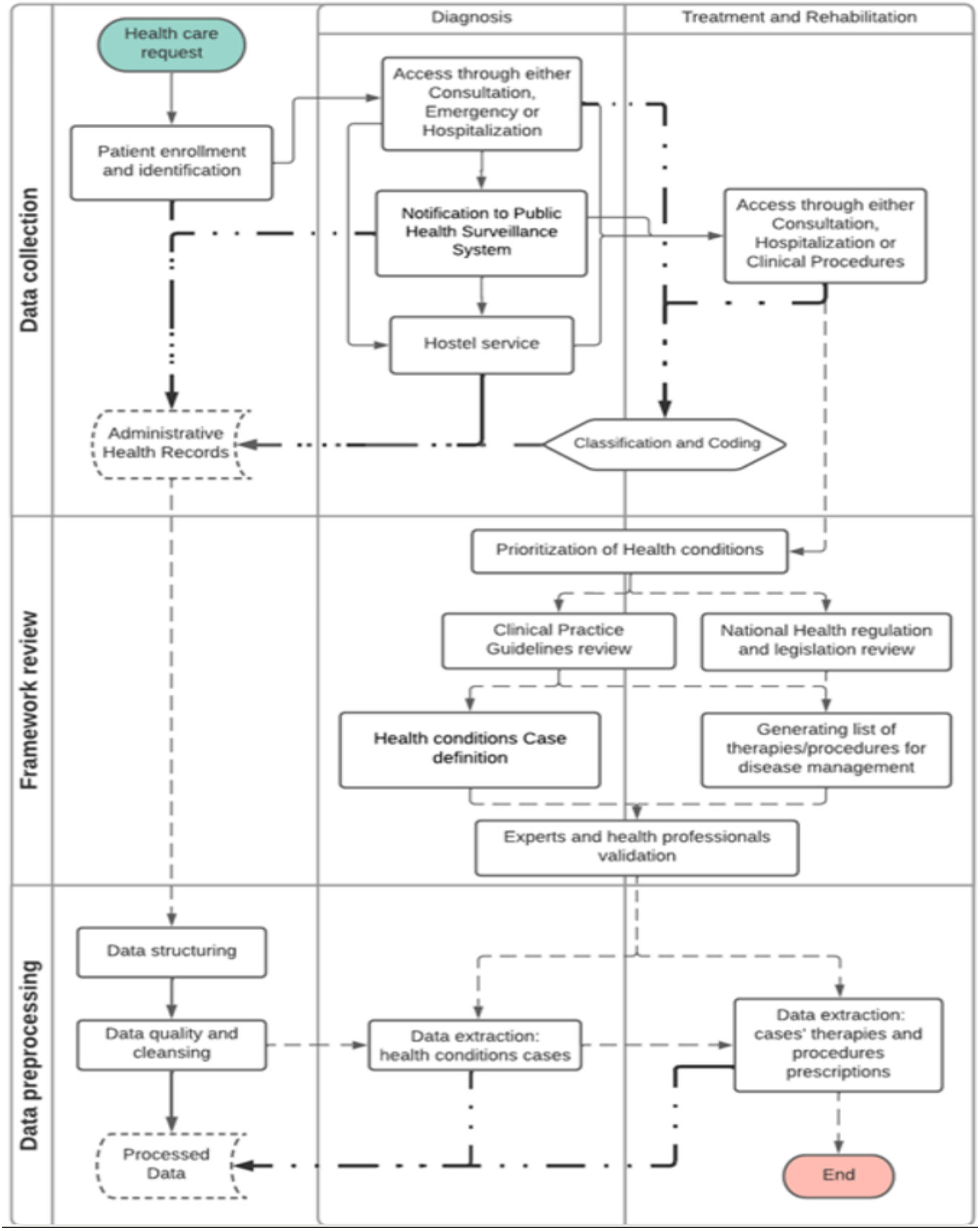


Figure 2. Study data collection, framework review and data preprocessing by healthcare stage. Solid line boxes identify actions performed. Dotted boxes represent outcomes. Solid lines show ways for Anas Wayuu enrollees to access health care services. Dotted lines show data flow. Dashed lines correspond to tasks carried out by the research team for data preprocessing.

for the adoption of data management paradigms that strengthen the public health decision-making process. This is consistent with the Colombian government's push for the integration and use of data for the analysis and monitoring of health outcomes, health policy, and social inequities [22]. Through the National Digital Strategy 2023-2026, the government promotes the development of infrastructure and protocols aimed at strengthening the availability, analysis, and exchange of data during public health decision-making [23]. These intentions are in line with international consensus that promotes the strengthening of data management to improve situation analysis and the provision of health, education and social welfare services to the population [3]. In a broader sense, these strategies align with the FAIR principles [16]. Indigenous health organizations in Colombia, while aligning themselves with these goals, have had to consider them in relation to the rights over their own data, as per the CARE principles [14]. While aspects of the principles can be somewhat complementary, our paper suggests a key tension.

Data governance in public health involves a systematic process of managing the phases of the health data lifecycle in order to produce quality information to inform public policy [1]. This includes designing data use policies, defining resources, maintaining infrastructure, designing and implementing data analytics processes, and identifying roles and human resources under internal processes for monitoring performance in data management, integrity, and confidentiality, as well as risk control. A data governance model considers three levels of involvement [2]. The first is an operational level with functions closely related to the data lifecycle (data quality, interoperability, confidentiality, and capacity building). The tactical level considers the design of participation strategies and alignment with national priorities, as well as the implementation of use cases. The strategic level, for its part, includes defining data management policies and processes, establishing the governance team supported by resources, and fostering multisectoral collaboration [2].

Health and healthcare monitoring, evaluation, and research present technical and methodological challenges for social actors involved in health management in their territories. Strengthening data governance is recognized as a path to reducing social inequities in health by progressively aligning processes for the custody, access, and use of public health data with communities. This involves engaging diverse social groups in analyzing the situation and managing public health knowledge within the population. Thus, the impetus for conceptualizing and implementing Indigenous data governance and sovereignty that has gained traction over the last 30 years, led by Indigenous organizations worldwide.

As Walter and Carroll [24] argue, up to recently Indigenous data have been used mostly by non-Indigenous policy makers to study Indigenous Peoples. They further state that Indigenous data sovereignty can invert the standard Indigenous data/policy nexus with Indigenous Peoples' own "governance of data and data for governance" [24]. Theodore and colleagues interpret this as meaning that "Indigenous Peoples govern and use their own data to inform their own policies and programs, and to monitor government and state policies that affect their communities" [25]. In consequence, Indigenous nations assert their claims to data sovereignty rooted in their inherent rights to self-determination as sovereign entities predating European settlers [26]. As suggested by Kukutai and Taylor, "Indigenous data sovereignty thus refers to the proper locus of authority over the management of data about Indigenous peoples, their territories and ways of life [26]. A corollary to these principles is the requirement to "build Indigenous expertise in the production and management of data and the formation of governance arrangements that allow for institutional oversight of research and data collection in Indigenous communities has been considered as one of the preconditions of data decolonization" [27].

Consistent with these principles, our signed agreement regarding the study's data management/structuring protocol affirmed the governance and sovereignty of the data by Anas Wayuu at the operational level (data quality, interoperability, confidentiality, and capacity building), at the tactical level (participation strategies, implementation of use), and at the strategic level (data management policies/processes, governance team, and multisectoral collaboration).There are similar experiences in other regions of the world, such as that of the Māori in New Zealand that have developed an administrative data infrastructure for future longitudinal studies as a contribution to Indigenous data sovereignty and Kaupapa Māori (Māori worldview) epidemiology [25].

A recent paper that thoroughly explored the complementarity and lack thereof between the FAIR principles and the CARE principles, concluded that the former "may not do justice to the data sovereignty of Indigenous Peoples as CARE principles intend" [28]. A clear example of the tension between both sets of principles in Anas Wayuu's case is that while as a health insurance organization it collects RIPS data from providers it contracts with, it must send a copy of the data to the Colombian Ministry of Health. Anas Wayuu also enters the data into its own information system and uses it for its purposes. In the case of our study, the data management protocol and data sharing agreement for research purposes demonstrated a clear instance of data governance and data sovereignty by Anas Wayuu. However, the fact that the data also ends up in the database of the Ministry of Health, while possibly in line with FAIR principles, seemingly curtails the rights of Indigenous Peoples right to data sovereignty. It is paramount that this contradictory tension be further explored and openly addressed.

Conclusion

The data management protocol for the use of administrative data of Anas Wayuu demonstrates two key aspects central to health research among Indigenous Peoples: the value of administrative data for health research; and that its management and use can be consistent with the principles of Indigenous data governance and sovereignty. To our knowledge, this is the first case from Latin America that has been published. The development and implementation of the protocol was somewhat aligned with the FAIR principles. However, Anas Wayuu being an Indigenous health organization had to consider them in relation to the rights over their own data, as per the CARE Principles for Indigenous Data Governance. While aspects of the two sets of principles can be somewhat complementary, our paper has suggested a key tension.

DECLARATIONS

AI utilization

Artificial intelligence tools such as data management and analysis software were solely used for code generation and the deployment of coded protocols aimed at exploring administrative data of the case study.

Competing interests

The authors declare that they have no competing interests.

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Author contributions

AH, BSA, DOG, HGV and JM co-designed the study. NMH validated data management protocol and supported data structuring. AH, DOG, FGC, YGC and YDB worked on data extraction, structuring, protocol documentation and data analysis. BSA, DOG, HGV and JM contributed to interpretation and drafted the manuscript.

Data availability

The raw/processed data required to reproduce the above findings cannot be shared publicly due to legal and ethical reasons.

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ABSTRACT IN SPANISH

Protocolo de gestión de datos para la investigación de servicios de salud utilizando datos administrativos: gobernanza y soberanía de los datos indígenas, Anas Wayuu, Colombia

La gobernanza de los datos de salud es un componente esencial de la gestión territorial de la atención sanitaria y del diseño de políticas públicas. Una gobernanza adecuada permite fortalecer las capacidades de gestión de datos a lo largo de todo su ciclo de vida. El análisis de la gobernanza de los datos de salud es particularmente relevante en relación con los pueblos indígenas, ámbito en el que el concepto de soberanía de los datos indígenas ha cobrado relevancia durante los últimos 30 años. Este artículo presenta un protocolo de gestión de datos desarrollado e implementado para un estudio de investigación por Anas Wayuu, una organización indígena de aseguramiento en salud de Colombia. Describimos y analizamos los fundamentos de los criterios técnicos utilizados para el procesamiento y la transferencia de datos administrativos, así como documentamos los elementos de un proceso de gestión de datos para su estructuración con el propósito de estudiar la utilización de los servicios de salud en el contexto de servicios de salud culturalmente adaptados.

Aunque el protocolo fue desarrollado e implementado en colaboración con socios académicos externos, mostramos cómo este proceso se llevó a cabo de manera coherente con los principios de gobernanza y soberanía de los datos indígenas. En cierta medida, el desarrollo y la implementación del protocolo estuvieron alineados con los principios FAIR (localizables, accesibles, interoperables y reutilizables). Sin embargo, al ser Anas Wayuu una organización indígena de salud, estos principios debían considerarse en relación con los derechos de los pueblos indígenas sobre sus propios datos, de acuerdo con los principios CARE para la gobernanza de datos indígenas. Si bien algunos aspectos de ambos conjuntos de principios pueden ser complementarios, nuestro artículo pone de manifiesto una tensión fundamental entre ellos.

Palabras clave: Soberanía de datos Indígena, gobernanza de datos, servicios de salud Indígena, Wayuu, CARE, FAIR.

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